Sihyeong Lee

AI-Native Full-Stack Product Engineer | Agentic Systems | TypeScript / Python | Production Integration

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US permanent resident, authorized to work without sponsorship. **Los Angeles based | Remote-first; open to LA hybrid/on-site work and U.S. domestic travel.**

Available immediately | Full-time or long-term contract

Summary

Engineer and former CTO with ten years of shipping software and nearly five years owning production systems and engineering outcomes. I use **Claude Code, Codex, and custom agents to accelerate implementation while retaining ownership of architecture, code review, testing, debugging, and release decisions**. My strongest work spans product, backend, infrastructure, and automation: turning ambiguous requirements into systems that are measurable, testable, and operable rather than treating generated code as the finished product.

Experience

Founder & Software Engineer, Soursea LLC | Los Angeles

Nov 2025 - Present | Los Angeles / Remote

Solo consultancy and product lab for AI integration and full-stack product work.

- Build **MCP (Model Context Protocol) servers and agent workflows** that connect LLMs to real systems: GPU generation backends, databases, project trackers, and deployment infrastructure, with typed tool calls, machine gates, and human approval where judgment should remain human.
- Use Claude Code, Codex, and isolated git-worktree sessions for parallel implementation while owning architecture, diff review, test strategy, failure analysis, and release decisions; validate with Playwright, Vitest, pytest, and task-specific evaluation harnesses.
- Build LLM features together with **retrieval benchmarks, deterministic validation, LLM-judge safety evals, and A/B tests** instead of treating model output as inherently correct.
- Operate PostgreSQL/pgvector, Docker, self-hosted GPU services, cloud VMs, Tailscale, Caddy, and Cloudflare across production and development environments.

Chief Technology Officer, Bitnara Inc. (ITEMBOX) | AI-assisted e-commerce platform

Jun 2024 - Oct 2025 | Remote

Owned the platform end to end and led a **nine-person distributed engineering team across Korea and Vietnam** while remaining its most active contributor.

- Owned a **NestJS/TypeORM backend with 20 API modules and roughly 300 REST endpoints** behind a Next.js frontend, covering multi-tenant workspaces, role-based permissions, team invitations, order workflows, proxy purchasing/quoting, and exchange-rate services.
- Replaced manual seller onboarding with **Shopify OAuth 2.0** and a Cafe24 integration; implemented HMAC validation, token storage, webhook registration, multi-currency stores, and cross-exchange-rate handling.
- Shipped **Stripe subscription billing** with renewal and payment-failure webhooks and operated the AWS footprint supporting the platform.
- Integrated image generation, background removal, translation, and OCR through **ComfyUI, ModelsLab, Kling, OpenAI, and Anthropic**, with FastAPI services, Redis/Celery workers, PostgreSQL, and MongoDB.

Chief Technology Officer, GEOMEC Inc.

Feb 2021 - May 2024 | Remote from the U.S. from 2023

Sole engineer responsible for internal software and infrastructure.

- Designed and shipped an **ElectronJS desktop client** with automated release, update, and logging pipelines.
- Built NAS-backed file services with role-based access, **Proxmox VE** virtualization, automated VM provisioning, and snapshot backups.
- Operated Linux servers, VPN, firewall rules, SSH key management, centralized logging, and internal network-security standards.

Quality Assurance, Case Farms Processing, Inc. | Morganton, NC

Mar 2023 - Mar 2024 | On-site

Worked directly on a U.S. manufacturing floor; started in production and **moved into quality assurance after five months**. First U.S. W-2 role in an English-language workplace while continuing GEOMEC CTO responsibilities remotely outside plant hours.

Independent Development & Study

2014 - 2020

Self-directed product work and occasional client projects while completing an associate degree and bachelor's degree.

Selected Projects

DevDeck | local-first command center for coding agents *(open source)*

TypeScript | Electron | React | Node.js | Vitest | Playwright

Command center for Claude Code, Codex, and Antigravity projects: repository/session state, embedded agent terminals, context and usage visibility, cross-project tasks, and direct machine-to-machine session control. Public CI currently reports **1,257 passing tests**; releases run across Windows, macOS, and Linux with automated multilingual, accessibility, and failure/recovery checks.

studios | multi-modal MCP platform on a shared kernel

Python | MCP | FastAPI | React | SQLite | ComfyUI | CUDA

Shared-kernel platform for image, video, 3D, voice, and music generation. The kernel centralizes the GPU backend adapter, GPU arbitration, asset storage, thumbnailing, and common MCP server skeleton. Tool output is machine-validated, then semantic judgment is persisted through a verify_asset workflow instead of disappearing inside a prompt.

HealFrame | AI-guided SaaS with a fail-closed safety pipeline

TypeScript | Next.js | Prisma/PostgreSQL | Google Gemini | Stripe | Capacitor

Solo full-stack web/Android product whose crisis-detection path is deliberately asymmetric: fail-closed on input, fail-open on output, with an evaluation harness used to test the safety boundary rather than relying on model confidence.

Technical Skills

AI / Agentic: Model Context Protocol (MCP), Claude API / Claude Code, OpenAI / Codex, Google Gemini, tool calling, agent orchestration, RAG, pgvector, cross-encoder reranking, LLM-judge evals, deterministic quality gates

Frontend: TypeScript, React, Next.js, Electron, React Native, Three.js, Tailwind CSS

Backend / Data: Node.js, NestJS, Python, FastAPI, PostgreSQL, MongoDB, Prisma, TypeORM, Redis, Celery, SQLite, PostGIS

Infra / Delivery: Docker, AWS (EC2, RDS, S3, Route 53, IAM), GitHub Actions, Proxmox VE, Playwright, Vitest, pytest, Tailscale, Caddy, Cloudflare, Vercel, PowerShell

Education & Languages

Hanyang Cyber University, B.S. in Computer Software Engineering, 2017-2019 | **Korea National Institute for Lifelong Education**, Associate degree in Computer Engineering, 2013-2016

Korean (native) | English (professional working)